Week 07: Programming and Research

Programming task

Session 2: deepen this week’s Python.
Predict → change → repair.
Session 3: apply it to the project.

Research task

Before Session 1: Research 5 warm-up.
After class: Research 6 on checking output.
Prepare your evidence; discuss in Week 8.

Open learning_materials/Week_07.ipynb. Follow Start Here; keep your work in that notebook. Email: submit.progmis@gmail.com.

Programming task

Session 2 · Python practice before project application.

1. Predict the exception path

Do this

Open #session-2: Predict → Run.
Start at `available = 5`; request 7.
Predict every printed line.

Observe and explain

Which line does raise skip?
Does the assignment complete?
Explain why the final count remains 5.

2. Change the request

Do this

Copy the example; request 2.
Then try 5 and 6 from a fresh 5.
Record messages and remaining count.

Observe and explain

2 → returns 3; 5 → returns 0.
6 → refusal; value remains 5.
Explain return versus raise.

3. Repair the handler placement

Do this

Run the SQLite interruption example.
The handler is inside with.
Predict the stored count.

Observe and explain

Move try/except outside with.
Rerun from setup: expect original 5.
Explain which block now sees the error.

Programming practice: done checklist

Keep in your notebook

Prediction before running
Your changed or repaired code
Observed output or error
One explanation of why it happened

Next step

Optional exploration can wait.
Save the notebook and a backup.
Continue to the project application.
Submit to submit.progmis@gmail.com.

Outside class: complete the practice and Research preparation.

Project application: complete reserve

Step	What to do / check
Open Session 3	#project-task: implement reserve(request_id, reviewed, lines, inject_failure=False).
Check input	Require review, known batch IDs and positive exact integer counts.
Change records together	Guard stock updates; save header and line unit snapshots in one transaction.
Keep the result	Capture offers, headers and lines with snapshot() before/after every case.

Use the supplied starter. Do not record collection or delivery.

Project checks: success and rollback

Step	What to do / check
Refuse without changes	Unreviewed Q11:1; later shortage Q11:1 + Q12:7; injected failure on a valid pair.
Success from opening 4/4	Reviewed Q11:2 + Q12:1 → stock 2/3, one header, two lines, 11 planned portions.
Additional checks	Zero, unknown batch after a valid line, repeated batch exhausting stock; explain a committed retry.
Save and explain	Your reserve tests, full snapshots, one peer finding and retest. Supplied practice tests are separate.

Research how to check automated or AI output

What to research

Find out how to verify an automated answer against its original source. You may use the supplied example instead of a live AI tool.

Possible directions

Compare fields, units, missing information and unsupported claims. Explain what you would accept, correct or ask about.

A short summary example

The source says “12 boxes offered”; the output says “12 portions collected”. The unit and collection claim are unsupported, so ask for clarification.

Discuss in Week 8 · summarize in your own words and cite your sources.